

1. Introduction

Stata offers several methods for imputation, including single imputation, multiple imputation, and interpolation. The choice of method depends on the nature of your data and the type of analysis you intend to perform. This guide provides step-by-step instructions for conducting multiple imputation of missing data using Stata.

Multiple imputation is one of the most robust and widely used statistical techniques for dealing with missing data. In multiple imputation, the distribution of observed data is used to estimate a set of plausible values for missing data. The missing values are replaced by the estimated plausible values to create a “complete” dataset.

2. Multiple Imputation in Stata

STEP 1: Preparing Your Data

1.1. Load your dataset. For this tutorial, we will use the "mheart5.dta", a data file available from Stata Corp. Type:

```
webuse "mheart5.dta"
```

1.2. Explore missing data. To examine the missing data pattern type:

```
misstable sum, gen(miss_)
```

The “misstable” command with the “gen()” option generates indicators for missingness. These new variables are added to the data file and start with the prefix miss_.

Stata will give us the following table.

						Obs<.
+-----						
					Unique	
Variable		Obs=.	Obs>.	Obs<.	values	Min
Max						
-----+-----+-----						

age		12		142	142	20.73613
83.78423						
bmi		28		126	126	17.22643
38.24214						

The `Obs=.` column represents the number of missing values for each variable. If there is no entry for a variable, it has no missing values.

The `Obs<.` column represents the number of observed values for each variable.

1.3. you may tabulate the new indicator variables as an additional check. Type:

```
tab1 miss_age miss_bmi
```

Stata will give us the following table.

```
-> tabulation of miss_age
```

(age>=.)	Freq.	Percent	Cum.
0	142	92.21	92.21
1	12	7.79	100.00
Total	154	100.00	

```
-> tabulation of miss_bmi
```

(bmi>=.)	Freq.	Percent	Cum.
0	126	81.82	81.82
1	28	18.18	100.00
Total	154	100.00	

Indicator variables `miss_age` and `miss_bmi` were added to the data file in step 1.2; a value of 1 on these variables indicates the observation is missing information on the

specific variable. A value of 0 indicates the observation is not missing. 12 observations are missing information on age, 28 observations are missing on BMI.

STEP 2: Consider the Mechanism of Missingness

Decide which variables need imputation. Not all missing data needs to be imputed; sometimes, missingness is informative. Multiple imputation is appropriate when data are missing completely at random (MCAR) or missing at random (MAR). It would be difficult to perform a legitimate analysis if data are missing not at random (MNAR).

2.1. Logistic regression models could be used to examine whether any of the variables in the data file predict missingness. If they do, the data are MAR rather than MCAR. To run the logit model, type:

```
logit miss_bmi attack smoke age female hsgrad
```

Stata will give us the following output table.

```
Iteration 0:  Log likelihood = -49.994502
```

```
Iteration 1:  Log likelihood =  -47.73123
```

```
Iteration 2:  Log likelihood = -47.614822
```

```
Iteration 3:  Log likelihood = -47.614515
```

```
Iteration 4:  Log likelihood = -47.614515
```

```
Logistic regression  
142
```

```
Number of obs =
```

4.76 LR chi2(5) =

0.4459 Prob > chi2 =

Log likelihood = -47.614515 Pseudo R2 =
0.0476

	miss_bmi	Coefficient	Std. err.	z	P> z	[95% conf. interval]
--	----------	-------------	-----------	---	------	-------------------------

-----+-----

attack		.0101071	.5775173	0.02	0.986	-1.121806
--------	--	----------	----------	------	-------	-----------

1.14202

smokes		.1965135	.5739319	0.34	0.732	-.9283723
--------	--	----------	----------	------	-------	-----------

1.321399

age		-.0485561	.0244407	-1.99	0.047	-.096459
-----	--	-----------	----------	-------	-------	----------

-.0006532

female		.0892789	.6256756	0.14	0.887	-1.137023
--------	--	----------	----------	------	-------	-----------

1.315581

hsgrad		.3940007	.6888223	0.57	0.567	-.9560662
--------	--	----------	----------	------	-------	-----------

1.744068

_cons		.1414761	1.423355	0.10	0.921	-2.648249
-------	--	----------	----------	------	-------	-----------

2.931201

age is statistically significantly associated with `miss_bmi`, suggesting that the data are MAR rather than MCAR.

2.2. Run another logistic model to make sure that no other variables other than BMI are statistically significantly associated with missingness of age. Type:

```
logit miss_age attack smoke female hsgrad
```

Stata will give us the following results.

```
Iteration 0:  Log likelihood = -42.144379
```

```
Iteration 1:  Log likelihood = -40.780233
```

```
Iteration 2:  Log likelihood = -40.713422
```

```
Iteration 3:  Log likelihood = -40.713172
```

```
Iteration 4:  Log likelihood = -40.713172
```

```
Logistic regression  
154
```

```
Number of obs =
```

```
2.86
```

```
LR chi2(4)    =
```

```
0.5811
```

```
Prob > chi2    =
```

```
Log likelihood = -40.713172  
0.0340
```

```
Pseudo R2     =
```

```
-----  
----
```

```
miss_age | Coefficient Std. err.      z    P>|z|    [95% conf.
interval]
```

```
-----+-----
-----

      attack |  -1.035628   .7108815   -1.46   0.145   -2.42893
.3576738

      smokes |   .2788896   .6369393    0.44   0.661   -.9694886
1.527268

      female |  -.0059384   .7025713   -0.01   0.993   -1.382953
1.371076

      hsgrad |   .5426292   .8029777    0.68   0.499   -1.031178
2.116437

      _cons |  -2.649692   .7993453   -3.31   0.001   -4.21638
-1.083004

-----+-----
-----
```

None of the p-values are less than 0.05 indicating no other variables are statistically significantly associated with missingness of age.

2.3. T-test may also be informative in evaluating whether the values of other variables vary between the missing and the non-missing groups. Type:

```
foreach var of varlist attack smoke age female hsgrad {
  ttest `var', by(miss_bmi)
}
```

Results for t test between miss_bmi and age are presented below. Results for t test for other variables with miss_bmi are not presented for brevity.

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.
interval]					
-----+-----					

0	126	57.14571	1.022929	11.48234	55.12121
59.17021					
1	16	50.82253	2.810969	11.24388	44.83109
56.81397					
-----+-----					

Combined	142	56.43324	.9727211	11.59131	54.51024
58.35624					
-----+-----					

diff		6.323186	3.040682		.3115936
12.33478					

diff = mean(0) - mean(1)				t =	
2.0795					

H0: diff = 0
140

Degrees of freedom =

Ha: diff < 0
0

Ha: diff != 0

Ha: diff >

Pr(T < t) = 0.9803
0.0197

Pr(|T| > |t|) = 0.0394

Pr(T > t) =

T-test suggests a statistically significant relationship between missingness of BMI and age. T-tests between missingness of BMI and the other variables (i.e., attack, smokes, female and hsgrad) were not statistically significant.

STEP 3: Setting-up Multiple Imputation

3.1: Declare MI set. Prior to imputation, data should be set to wide using the “mi set” command. This set up suggests Stata how the additional imputations should be stored.
Type:

```
mi set wide
```

3.2. Register variables for imputation. Variables in the data set have to be registered using the “mi register” command. “mi register regular” specifies the variables that should not be imputed (either because they have no missing values or because there is no need). “mi register imputed” specifies the variables to be imputed in the procedure.
Type:

```
mi register regular female attack smokes hsgrad  
mi register imputed bmi age
```

STEP 4: Choosing an Imputation Method

- Choose an appropriate method for each type of variable you want to impute. For instance, choose `regress` for continuous variables (e.g., age, bmi), `logit` for binary variables (e.g., gender), `mlogit` for multinomial variable (e.g., political parties).
- Stata offers `mi impute chained` (MICE) for a flexible approach that can handle different types of variables (e.g., continuous, binary, ordinal, nominal, truncated, count) in one command.
- `mi impute chained` is an iterative process. The variable with the fewest missing values is imputed first followed by the variable with the next fewest missing values and so on for the rest of the variables.

STEP 5: Imputing the Missing Data

5.1. Perform the imputation. To impute missing values using chained equations for 5 imputations, type:

```
mi impute chained (regress) age bmi = attack smokes hsgrad female, add(5)
rseed(9478) replace
```

Interpretation of the codes:

`mi impute chained`: command for chained imputation with a mixed varieties of variables.

(regress) age bmi: Both BMI and age, the variables to be imputed, are continuous variables. Therefore, regress is specified as a method.

attack smokes hsgrad female: These are regular variables which should not be imputed.

add(5) : The number of datasets to be imputed (5).

rseed(9478) : The “rseed()” option may be used for results reproducibility.

Stata will give us the following table.

note: missing-value pattern is monotone; no iteration performed.

Conditional models (monotone):

age: regress age attack smokes hsgrad female

bmi: regress bmi age attack smokes hsgrad female

Performing chained iterations ...

Multivariate imputation	Imputations =	5
-------------------------	---------------	---

Chained equations	added =	5
-------------------	---------	---

Imputed: m=1 through m=5	updated =	0
--------------------------	-----------	---

Initialization: monotone	Iterations =	0
--------------------------	--------------	---

burn-in =	0
-----------	---

```
age: linear regression
```

```
bmi: linear regression
```

Observations per m				
Variable	Complete	Incomplete	Imputed	Total
age	142	12	12	154
bmi	126	28	28	154

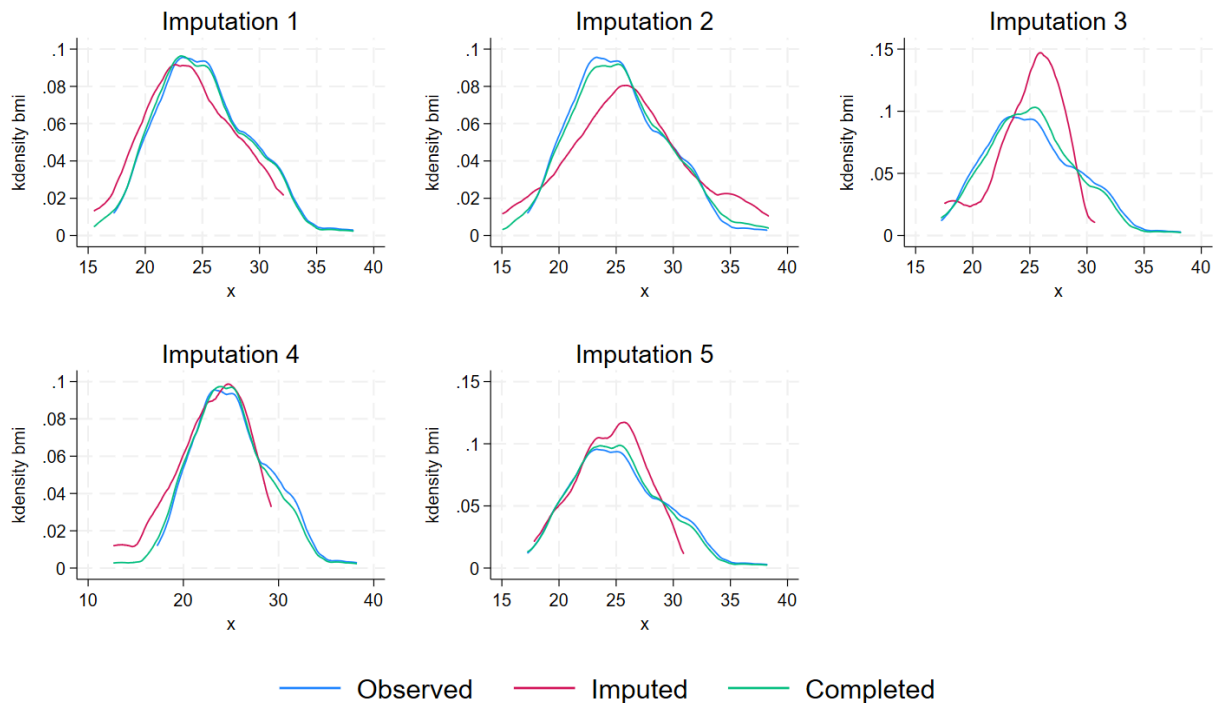
(Complete + Incomplete = Total; Imputed is the minimum across m of the number of filled-in observations.)

`Imputations = 5` and `added = 5` indicate five imputations of variables which have missing values on the observed dataset were added. The new variables are noted with a prefix `_x_` where x represent the imputation number (i.e. , 1, 2,...,5).

5.2. Check the imputation. To compare the distribution of imputed variables to that of the observed and the complete (i.e., observed and imputed) data use the `middiagplots` command which can be downloaded by typing `findit middiagplots` in the command line. In sum, type the following codes:

```
findit midiagplots
midiagplots bmi, m(1/5) combine
```

Stata will give us the following plots.



The above plots represent the distribution of BMI (this could be done for any of the imputed variables) and suggest a good overlap between observed and completed data.

STEP 6: Analyzing the Imputed Data

6.1. Perform your analysis. Use `mi estimate` to perform your desired statistical analysis on the imputed dataset. For example, to run a logit model, you would use:

```
mi estimate, or: logit attack bmi age female smokes hsgrad
```

Interpretation:

In the above codes, we use a logistic regression to estimate the probability of a heart attack with the pooled 5 imputed data files.

The “mi estimate” prefix first runs the estimation command on each of the imputations separately. It then combines the results and displays the combined output.

Using the “or” option will present odds ratios following a logistic regression.

Stata gives us the following output table:

Multiple-imputation estimates	Imputations	=
5		
Logistic regression	Number of obs	=
154		
	Average RVI	=
0.0454		
	Largest FMI	=
0.1607		
DF adjustment: Large sample	DF: min	=
175.09		
	avg	=
2243117.63		
	max	=
1.30e+07		

Model F test: Equal FMI F(5, 5615.2) =
3.13

Within VCE type: OIM Prob > F =
0.0080

	attack	Odds ratio	Std. err.	t	P> t	[95% conf. interval]
--	--------	------------	-----------	---	------	-------------------------

-----+-----

bmi	1.094327	.0525146	1.88	0.062	.9954409
-----	----------	----------	------	-------	----------

1.203037

age	1.028546	.016244	1.78	0.075	.997173
-----	----------	---------	------	-------	---------

1.060906

female	.8829223	.3608274	-0.30	0.761	.3963305
--------	----------	----------	-------	-------	----------

1.966924

smokes	3.138241	1.100922	3.26	0.001	1.577892
--------	----------	----------	------	-------	----------

6.241591

hsgrad	1.160312	.4648238	0.37	0.711	.5291435
--------	----------	----------	------	-------	----------

2.544345

_cons	.0097271	.0156492	-2.88	0.004	.0004141
-------	----------	----------	-------	-------	----------

.2284712

Note: _cons estimates baseline odds.